SEC 3 · LEC 1 · ORBITAL MECHANICS

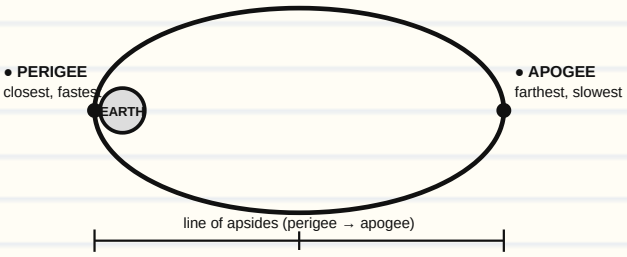
# Six numbers pin down any orbit

Left = your left notebook page. Right = your right page. Copy what's here; draw the pictures yourself before you look closely at mine.

## LEFT PAGE · DRAW

**S3 / L1 — Orbit parameters** **date · pg __**

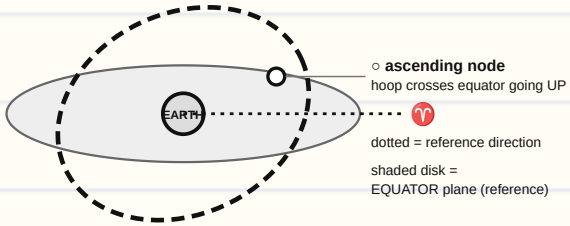
**1 · THE ELLIPSE: Earth sits OFF-CENTER (at a focus)**




**a = (r_perigee + r_apogee) ÷ 2 → SIZE · half of this line**

e = how squashed: 0 circle · 0.5 egg · near 1 cigar · over 1 = open path (flyby)
both radii measured from Earth's CENTER, not the surface

**2 · THE TILTED HOOP: i, Ω, ω**




**i** = TILT of the hoop vs the equator
**Ω** = SWIVEL: angle from ♈ around to the ascending node
**ω** = SPIN in plane: angle from node to perigee · **ν** = where it is NOW

✏️ **PENCIL KEY** · solid = the orbit · dashed = the tilted hoop · shaded disk = equator plane · ● filled dot = perigee / apogee · ○ hollow dot = ascending node · dotted line = ♈ reference direction

⭐ **3 TAKEAWAYS** 3 lines

1. a = size, e = shape, i / Ω / ω = how the hoop is turned, ν = where the bead is
2. Perigee and apogee are measured from Earth's center; a is their average
3. Reference for Earth orbits = equator plane + ♈ direction (NOT ecliptic / Sun)

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

## RIGHT PAGE · CORNELL

| CUES (questions) | NOTES (keywords) |
|---|---|
| Orbital mechanics in one line? | orbital mechanics = the math of where a spacecraft goes, given its position + speed when the rocket cuts off |
| a = ? from what two points, measured from where? | 2 jobs: orbits around Earth · transfers Earth → another planet |
| e = 0 is what? 0.7? more than 1? | perigee = closest point · apogee = farthest · both from Earth's CENTER |
| Where are you fastest / slowest? Where do you fire the engine to change the orbit? | perigee = fastest, apogee = slowest · burn at perigee → apogee goes up · burn at apogee → perigee goes up |
| Which 5 numbers freeze the orbit, which 1 moves? | a = (r_p + r_a) ÷ 2 → SIZE · for a circle, a = r |
| Hula hoop: which one is tilt, swivel, spin-in-place? | e = squash: 0 circle · 0.5 egg · near 1 cigar · over 1 = open path (flyby) |
| Reference plane + direction for an Earth orbit? | i = tilt vs equator: 0° = equatorial · 90° = polar · ISS = 51.6° |
| Same Ω, different ω: what moved? | Ω = swivel: where the orbit crosses the equator going UP ( ascending node ), measured from ♈ |
| ? what e does a flyby have | ω = spin in-plane: angle from node → perigee |
| | ν = where the s/c is NOW · the only one that changes minute to minute |
| | ref plane = EQUATOR · ref direction = ♈ (instructor said ecliptic / "toward the Sun": wrong for Earth orbits) |

SEC 3 · LEC 1 · TEACH + FACT CHECK

# The hula hoop, one parameter at a time

Start with the box. If the box makes sense, everything under it is just adding names to things you already get.

## 🧒 Like you're 5

You're on a playground and you throw a ball sideways, really hard. It falls, but it also goes forward, so it lands far away. Now throw it so hard that by the time it falls, the ground has curved away underneath it. It never lands. It keeps falling *around* the Earth forever. That's an orbit: **falling and missing.**

Now picture the ball's path as a **hula hoop around a basketball**. To tell a friend exactly how your hoop sits, you'd say six things: How big is it? How squished? How tilted? Which way does the tilt face? Is the squished end pointing up or sideways? And where on the hoop is the ball right now? That's all the six fancy names below are. **Six ways to describe a hoop.**

## 🏀 Basketball + hula hoop

Earth is the basketball. The orbit is a hoop around it. To describe exactly how the hoop sits, you need six things:

- **a, semi-major axis: how big.** Half the long way across the ellipse. Equivalently the average of the closest distance (perigee) and farthest distance (apogee), both measured from Earth's center. Bigger a = bigger orbit = longer period (Kepler's 3rd law, coming later).
- **e, eccentricity: how squashed.** 0 is a perfect circle. 0.3 is a gentle egg. 0.9 is a cigar with Earth near one tip. Exactly 1 means the path never closes (escape). More than 1 is a flyby: the spacecraft swings past and leaves. Three orbits with the same e but different a look like the same shape at three zoom levels. That's the left picture on the concepts slide.
- **i, inclination: how tilted.** The angle between the hoop and Earth's equator. 0° = hoop lies flat on the equator (GEO sats). 90° = hoop passes over both poles (polar orbit). 51.6° = the ISS.
- **Ω, longitude of the ascending node: swivel.** A tilted hoop crosses the equator at two points. The one where the satellite is heading north is the **ascending node**. Ω says how far around the equator that crossing point is, measured from a fixed starting direction. Keep the tilt, swivel the hoop around Earth's axis: Ω changes, nothing else does. That's the top-right picture: three identical tilted orbits swiveled to different Ω.
- **ω, argument of periapsis: spin in place.** Keep the hoop exactly where it is and rotate it within its own plane, so the squashed end (perigee) points somewhere else. Measured from the ascending node to perigee. The middle and bottom-right pictures: same node, same shape, perigee rotated.
- **ν, true anomaly: where the bead is.** The other five freeze the hoop in space. ν is the angle from perigee to where the spacecraft is right now. It's the only one that changes every second.

## 🔎 Caption errors (this transcript was rough)

| Captions said | Instructor meant |
|---|---|
| "radius of perjury" | radius of **perigee** |
| "Perry axis", "paradoxes", "Perry obsessed", "UPS's" | **periapsis** (= perigee, general word) |
| "the cryptic plane" | the **ecliptic** plane |
| "your mega" | **omega**, Ω |
| "lunches do for sending note" | longitude of ascending node |
| "DMA major axis", "some major accident" | **semi-major** axis |
| "IC centric" | **eccentric** |
| "two anomaly" | **true** anomaly |

## ▲ The instructor's real mistake

- **He said the reference plane is "the plane containing Earth and the Sun, the ecliptic," and the reference direction "points from Earth to the Sun."** He hedged with "I believe," and he's wrong for Earth orbits. For a satellite around Earth, the reference plane is **Earth's equator**, and the reference direction is the **vernal equinox** (the ♈ symbol on his own slide): where the Sun sits on the first day of spring, fixed against the stars. "Toward the Sun" moves every day, so it can't be a reference. The ecliptic is the reference for orbits around the *Sun*. Write the equator version.
- **Minor:** "semi-major axis is the distance from one point of the orbit to the other divided by two." Only if those two points are perigee and apogee. The formula on your page is the precise one.
- **Correct and worth noting:** e runs 0 to 1 for closed orbits. He stopped there; the e ≥ 1 cases are escape and flyby, which is

SEC 3 · LEC 1 · VOCABULARY

# Every word used, in plain English

Lots of Greek letters this time. They're just labels; the meaning column is what matters. Copy only the new ones into the word bank.

**Orbital mechanics** — the math of how things move under gravity: given where a spacecraft is and how fast it's going, where will it be later? *Ballistics, but the "ground" is a planet.*

**Transfer orbit** — an orbit you use to get from one place to another, like Earth to Mars, rather than to stay. *The highway, not the parking lot.*

**Semi-major axis (a)** — half the longest width of the ellipse. Sets the orbit's size and its period. *"Semi" = half, "major axis" = the long way across.*

**Perigee / apogee** — closest and farthest points of an orbit around Earth, measured from Earth's center. *"Peri" = near, "apo" = far, "gee" = Earth.*

**Apsis (plural apsides)** — either of those two extreme points. The line joining them is the long axis of the ellipse.

**Equator / equatorial plane** — the flat sheet through Earth's middle, perpendicular to its spin axis. The reference plane for Earth orbits.

**Plane of reference** — the flat sheet you measure tilt from. Equator for Earth orbits, ecliptic for Sun orbits.

**Node / ascending node** — the two points where a tilted orbit pierces the reference plane. The ascending node is the one where the satellite is crossing from south to north. *Where the hoop pokes up through the table.*

**Argument of periapsis (ω, small omega)** — the angle, measured in the orbit's own plane, from the ascending node to perigee. The orbit's spin-in-place. *"Argument" is old math-speak for "angle." Nobody is arguing.*

**Greek letters** — Ω omega (big), ω omega (small), ν nu. Just labels, like x and y. Say them out loud once so they stop looking scary.

**Polar orbit / equatorial orbit** — i = 90° (passes over both poles, sees the whole Earth as it spins below) / i = 0° (stays over the equator).

**Constraints / requirements** — what the mission needs (constraints: limits you can't break; requirements: things it must do). The orbit is chosen to satisfy them.

**Celestial body** — any big natural object in space: planet, moon, Sun, asteroid, comet. The thing you orbit.

**Orbital parameters / elements** — the six numbers that fully describe an orbit: a, e, i, Ω, ω, ν. *An orbit's address.*

**Eccentricity (e)** — a number from 0 up that says how squashed the ellipse is. 0 = circle, near 1 = very stretched, 1 or more = doesn't close. *"Eccentric" literally means off-center.*

**Periapsis / apoapsis** — the same two points, generic word for any body. Perihelion/aphelion = around the Sun. Perilune/apolune = around the Moon. *Same idea, different suffix per planet.*

**Inclination (i)** — the tilt angle between the orbit's plane and the reference plane (the equator, for Earth). 0° flat, 90° over the poles.

**Ecliptic** — the flat sheet Earth's orbit around the Sun lies in. The reference plane for orbits around the Sun. *Not the same as the equator; they're tilted 23.4° apart.*

**Reference direction / vernal equinox (♈)** — a fixed "12 o'clock" direction in space to measure angles from: the direction to the Sun on the first day of spring. The ♈ symbol (Aries) marks it on diagrams. *North on a compass, but for space.*

**Longitude of the ascending node (Ω, capital omega)** — the angle, measured in the reference plane from ♈, to the ascending node. The orbit's swivel. *Which direction the hoop's "up-crossing" points.*

**True anomaly (ν, nu)** — the angle from perigee to where the spacecraft is right now. *"Anomaly" is old astronomy-speak for "angle along the orbit." Nothing is wrong.*

**Circular / elliptical / hyperbolic** — orbit shapes by e: circle (e = 0), ellipse (0 < e < 1), hyperbola (e > 1, an open swing-past like a flyby).

**Projection** — looking at a 3-D orbit flattened onto a 2-D page, like a shadow. Why the slide pictures can be hard to read.

**Geometry (of an orbit)** — its size, shape and orientation: a, e, i, Ω, ω. Everything except where the spacecraft currently is.

SEC 3 · LEC 1 · SAY IT OUT LOUD

# Page 4: prove it by talking

Do this AFTER the spread is written and the 3-line summary is done, with the notebook and this PDF closed. Send me what you said (typed or dictated) and I fill the box at the bottom.

## 🗣 The friend's opening question

"You're taking a spacecraft course? What did you learn about orbits?"

## ✅ Three sentences you should be able to say

- An orbit is falling and missing: you go sideways so fast the ground curves away under you.
- You can describe any orbit with six numbers: how big, how squished, how tilted, which way the tilt faces, which way the squished end points, and where the spacecraft is right now.
- The closest point is perigee, the farthest is apogee, and the size of the orbit is basically the average of the two.

No Greek letters required. If a friend asks for the names, that's a bonus, not the test.

## ❓ Follow-ups a friend might ask

- "Wait, why doesn't it just fall down?"
- "So is the ISS going in a circle or an oval?" (nearly circular, e ≈ 0)
- "What's the difference between tilt and which-way-the-tilt-faces?" (inclination vs Ω: tilt the hoop, then swivel it)
- "Why six numbers and not three?" (three for the shape and size won't tell you how it's turned in space)
- "If you only changed one number, which one changes every second?" (where the spacecraft is: true anomaly)

## 📝 Your explanation (paste or dictate)

_ send it to me in chat; I'll put it here _

## 🎯 My read

solid on: _
fuzzy on: _
fix: _
→ question-bank cards: _

**⚡ Rule from here on: a lecture isn't "done" until this page has something in both boxes.**